

NOTE:



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1. Calculate the polynomials $(12x^4 - 5x + 18 + 6x^4) + (13x^2 + 7x - 9)$

- ☐ A. $18x^4 + 8x^2 + 25x - 9$
☐ B. $18x^8 + 8x^3 + 25x - 9$
☐ C. $18x^8 + 13x^3 + 2x + 9$
☐ D. $18x^4 + 13x^2 + 2x + 9$

2. What is the difference between $2x^2 + 3x - 2$ and $5x^2 - x - 7$?

- ☐ A. $-3x^2 + 4x + 5$
☐ B. $3x^2 + 4x + 5$
☐ C. $7x^2 + 4x + 9$
☐ D. $-3x^2 + 2x + 9$

3. Determine $(2x^5 + 3x^2) - (x^5 - 7x^2)$.

- ☐ A. $x^{10} + 10x^4$
☐ B. $x^5 - 4x^2$
☐ C. $3x^5 - 4x^2$
☐ D. $x^5 + 10x^2$

4. Which polynomial is equivalent to $(x^2 + 7)(12x^3 - 6)$?

- ☐ A. $12x^5 + 84x^3 - 6x^2 - 42$
☐ B. $12x^3 + x^2 + 1$
☐ C. $12x^6 + 84x^3 - 6x^2 - 42$
☐ D. $12x^6 - 42$

5. Which expression is equivalent to $2x^5(x^3 + 5x)$?

- ☐ A. $3x^8 + 7x^6$
☐ B. $2x^{15} + 10x^5$
☐ C. $2x^{15} + 7x^5$
☐ D. $2x^8 + 10x^6$

6. Simplify the expression $a^2b^3(a^4b^4)$

☐ A. a^5b^{12}

☐ B. a^6b^{12}

☐ C. a^6b^7

☐ D. a^5b^7

7. Convert the expression $\sqrt{ab}$, where a and b are positive, into rational exponent notation.

☐ A. $a^{1/2}b$

☐ B. $ab^{1/2}$

☐ C. $a^{1/2}b^{1/2}$

☐ D. ab^2

8. Which of the expressions is equivalent to $g^{\frac{2}{5}}h^{\frac{4}{5}}$?

☐ A. $\sqrt[5]{g^2h^4}$

☐ B. $\frac{1}{\sqrt[4]{g^5h^{10}}}$

☐ C. $\sqrt[4]{g^5h^2}$

☐ D. $\frac{1}{\sqrt[5]{g^4h^2}}$

9. Apply the product rule and power rule to simplify $k^{\frac{5}{16}}\left(k^{\frac{3}{2}}\right)^{\frac{5}{8}}$ where $k > 0$.

☐ A. $\sqrt{k}$

☐ B. $\sqrt[4]{k^5}$

☐ C. $\sqrt[8]{k^5}$

☐ D. $\sqrt[15]{k^{16}}$

10. Which of the expressions is equivalent to $y^{\frac{1}{8}} \left(y^{\frac{3}{4}} \right)^{\frac{3}{2}}$

- ☐ A. $\sqrt{y}$
- ☐ B. $\sqrt[4]{y^5}$
- ☐ C. $\sqrt[8]{y^5}$
- ☐ D. $\sqrt[8]{y^7}$

11. Which of the following is equivalent to the expression $(\sqrt{2p} - \sqrt{2r})^{\frac{2}{3}}$ where $x > y$ and $y > 0$

- ☐ A. $(2q + 2r)^5$
- ☐ B. $(2q - 2\sqrt{qr} + 2r)^{\frac{1}{5}}$
- ☐ C. $\sqrt[3]{2q + 2r}$
- ☐ D. $\sqrt[3]{2q - 4\sqrt{qr} + r}$

12. Which of the following is equivalent to the expression $(2\sqrt{x} - \sqrt{y})^{\frac{2}{5}}$ where $x > y$ and $y > 0$

- ☐ A. $(4x - y)^5$
- ☐ B. $(4x - 4\sqrt{xy} + y)^{\frac{1}{5}}$
- ☐ C. $\sqrt[5]{4x - y}$
- ☐ D. $\sqrt[5]{4x - 4xy + y}$

13. If $\sqrt[3]{a^2} = \sqrt{b}$, $a^{2x} = b^6$ where a and b are constant $a > 1$ and $b > 1$, what is the value of x ?

- ☐ A. 2
- ☐ B. 3
- ☐ C. 4
- D. 12

14. Two numbers, a and b , are each greater than zero, and the square root of a is equal to the cubic root of b . For what value of x is $a^{(2x-1)}$ equal to b ?

- ☐ A. 1
- ☐ B. $\frac{3}{2}$
- ☐ C. $\frac{5}{4}$
- ☐ D. 3

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